Ranking Flaky Test Failures

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***Abstract*—Flaky tests pass and fail on the same code. Which failure is flaky? The ranking answers this question.**

I. Introduction

Flaky tests pass and fail on the same code [**?**]. Detectors rerun each failure [**?**], [**?**], and a survey covers the field [**?**]. Section **??** describes the method, Fig. **??** gives an overview, Table **??** lists the results, and Eq. (**??**) defines the score, also written as (**??**). Algorithm **??** ranks the failures, RQ**??** asks how accurate the ranking is, and **??** and **??** close the paper on page **??**.

A developer asked, "Why?? The test passed yesterday!" Another wrote "Really?!" and a third asked, "Is this the right page??" The value $f(?)$ stays open, and the most common (?) cause is the order of the tests. Is the cause known? Not yet. The ternary form `a ? b : c` and the query `?q=flaky` appear in the logs.

1) **RQ1:** How accurate is the ranking? (**RQ1**)

$$s(t) = \frac{f(t)}{r(t)} \quad (?) \tag{1}$$

II. Method

What now?? The tool ranks the failures. **Is it flaky?** The history decides.

```
grep -E "[?]" log.txt   # a literal ?? or (?) in the log
```

```
var name = user?.Name ?? "unknown";
```

TABLE I
Results

Project	Accuracy	Flaky?
A	0.71	yes
B	?	(?)



Fig. 1. Overview

Algorithm 1 Rank

rank the failures

III. Discussion

Which failures need attention first? The ranking answers this question for 71% of the builds.